

Logistic Regression

1. Introduction

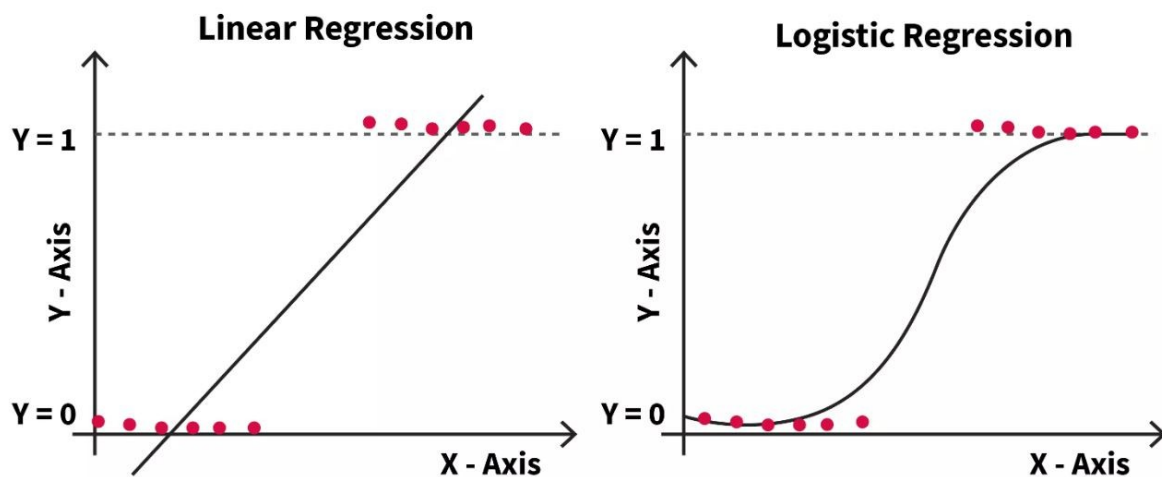
Logistic Regression is a supervised learning algorithm used for **binary classification** problems where the target variable has two possible outcomes (0 or 1, Yes or No, True or False).

Despite its name containing "regression," it's a **classification algorithm** that predicts the probability of an instance belonging to a particular class.

2. Why Not Linear Regression for Classification?

Linear regression outputs continuous values from $-\infty$ to $+\infty$, which doesn't work well for classification because:

- Probabilities must be between 0 and 1
 - Linear regression is sensitive to outliers
 - It doesn't provide a natural decision boundary
-



3. The Logistic Function (Sigmoid Function)

Logistic regression uses the **sigmoid function** to map predictions to probabilities:

Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Where:

- $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$ (linear combination of features)
- e = Euler's number (≈ 2.718)

Properties:

- Output range: (0, 1)
 - S-shaped curve
 - When $z=0$, $\sigma(z) = 0.5$
 - As $z \rightarrow \infty$, $\sigma(z) \rightarrow 1$
 - As $z \rightarrow -\infty$, $\sigma(z) \rightarrow 0$
-

4. Mathematical Foundation

4.1 Hypothesis Function

$$h_{\theta}(x) = \sigma(z) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

4.2 Decision Boundary

- If $h_{\theta}(x) \geq 0.5$, predict class 1
- If $h_{\theta}(x) < 0.5$, predict class 0

The threshold of 0.5 is commonly used but can be adjusted based on the problem.

4.3 Odds and Log-Odds

Odds:

$$\text{Odds} = \frac{p}{1 - p}$$

Where p is the probability of success.

Log-Odds (Logit):

$$\log\left(\frac{p}{1 - p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

This shows that logistic regression models the log-odds as a linear function of features.

5. Cost Function (Log Loss)

Linear regression's MSE doesn't work for logistic regression because it creates a **non-convex** function with multiple local minima.

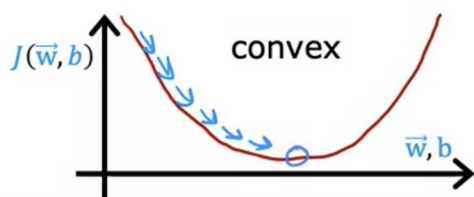
Squared error cost

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m \left[\frac{1}{2} (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2 \right]$$

loss $L(f_{\vec{w}, b}(\vec{x}^{(i)}), y^{(i)})$

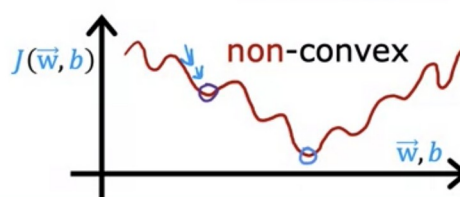
linear regression

$$f_{\vec{w}, b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$



logistic regression

$$f_{\vec{w}, b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$



5.1 Binary Cross-Entropy (Log Loss)

$$\text{Cost}(h_{\theta}(x), y) = -[y \cdot \log(h_{\theta}(x)) + (1 - y) \cdot \log(1 - h_{\theta}(x))]$$

For m training examples:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y_i \cdot \log(h_{\theta}(x_i)) + (1 - y_i) \cdot \log(1 - h_{\theta}(x_i))]$$

Why this cost function?

- Convex function (single global minimum)
 - Penalizes confident wrong predictions heavily
 - Derived from maximum likelihood estimation
-

6. Optimization: Gradient Descent

We minimize the cost function using gradient descent.

6.1 Update Rule

$$\theta_j := \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j}$$

6.2 Gradient Calculation

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m \left[(h_{\theta}(x_i) - y_i) \cdot x_{ij} \right]$$

Where:

- α = learning rate
- m = number of training examples
- x_{ij} = jth feature of ith example

Note: The gradient formula looks identical to linear regression, but $h_{\theta}(x)$ is different (sigmoid vs linear).

7. Types of Logistic Regression

7.1 Binary Logistic Regression

- Two possible outcomes (0 or 1)
- Example: Spam vs Not Spam

7.2 Multinomial Logistic Regression

- More than two unordered classes
- Example: Predicting cat, dog, or bird
- Uses **softmax function** instead of sigmoid

7.3 Ordinal Logistic Regression

- More than two ordered classes
 - Example: Rating (Poor, Average, Good, Excellent)
-

8. Assumptions

1. **Binary or ordinal dependent variable**
 2. **Independence of observations:** No autocorrelation between observations
 3. **Little to no multicollinearity:** Independent variables shouldn't be highly correlated
 4. **Linearity of log-odds:** Linear relationship between independent variables and log-odds
 5. **Large sample size:** Generally requires larger samples than linear regression
 6. **No extreme outliers:** Can significantly affect the model
-

9. Model Evaluation Metrics

9.1 Confusion Matrix

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	True Negative (TN)	False Positive (FP)
Actual Positive (1)	False Negative (FN)	True Positive (TP)

9.2 Key Metrics

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision (Positive Predictive Value):

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Of all positive predictions, how many were correct?

Recall (Sensitivity, True Positive Rate):

$$\text{Recall} = \frac{TP}{TP + FN}$$

- Of all actual positives, how many did we catch?

Specificity (True Negative Rate):

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- Of all actual negatives, how many did we identify correctly?

F1-Score:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Harmonic mean of precision and recall

9.3 ROC and AUC

ROC Curve (Receiver Operating Characteristic):

- Plots True Positive Rate vs False Positive Rate at various thresholds

AUC (Area Under ROC Curve):

- 0.5 = random guessing
- 1.0 = perfect classifier

- Generally: 0.7-0.8 = acceptable, 0.8-0.9 = excellent, >0.9 = outstanding
-

OVR (One-vs-Rest)

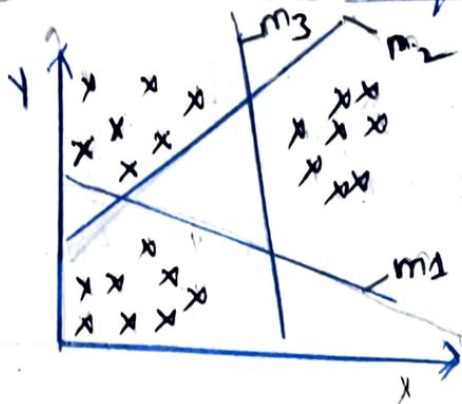
Definition:

OVR (One-vs-Rest) is a strategy used to extend logistic regression (which is naturally a binary classifier) to multi-class classification problems.

Explanation:

- In OVR, the algorithm trains one classifier per class.
- Each classifier treats one class as the positive class (1) and all the remaining classes as the negative class (0).
- During prediction, the model computes the probability of each class, and the class with the highest probability is chosen as the final prediction.

⑤ multiclass classification { Logistic Regression }



$m_1 \rightarrow$ Binary classification

$m_2 \rightarrow$ Binary classification

$m_3 \rightarrow y$

One versus Rest (OVR) $\rightarrow$ Logistic Regression.

f_1	f_2	f_3	$op \rightarrow 0_1, 0_2, 0_3$	one hot encoding		
				0_1	0_2	0_3
-	-	-	0_1	1	0	0
-	-	-	0_2	0	1	0
-	-	-	0_3	0	0	1
-	-	-	0_1	1	0	0
-	-	-	0_3	0	0	1
-	-	-	0_2	0	1	0

$\rightarrow$ with the help of ovr we try to create internally

multiple models

$\rightarrow$ Each and Every model act as binary classification

model

$$M_1 \leftarrow \mathcal{D}_p \{f_1, f_2, f_3\} \{0_1\}$$

$$M_2 \leftarrow \mathcal{D}_p \{f_1, f_2, f_3\} \{0_2\}$$

